



Real-time electrocardiogram P-QRS-T detection–delineation algorithm based on quality-supported analysis of characteristic templates



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ABSTRACT

The main objective of this study is to introduce a simple, low-latency, and accurate algorithm for real-time detection of P-QRS-T waves in the electrocardiogram (ECG) signal. In the proposed method, real-time signal preprocessing, which includes high frequency noise filtering and baseline wander reduction, is performed by applying discrete wavelet transform (DWT). A method based on signal first-order derivative and adaptive threshold adjustment is employed for real-time detection of the QRS complex. Moreover, detection and delineation of P- and T-waves are achieved by correlation analysis conducted between signal and their templates. Besides, signal quality is investigated online, and if the quality of the analysis window is unacceptable, then the algorithm will guess (estimate) the locations of P- and T-waves.

The operating characteristics of the proposed algorithm are evaluated by its implementation to an artificially generated ECG signal whose quality is adjustable from the best (Quality, 100%) to the worst (Quality, $\leq 40\%$) cases based on the random-walk noise theory. The algorithm was applied to the MIT-BIH arrhythmia database, QT database, and Physionet/CinC challenge 2011 competition database. The obtained results, which were based on the QT database, showed sensitivity and positive predictivity of $Se = 99.63\%$ and $P+ = 99.83\%$, $Se = 99.83\%$ and $P+ = 99.98\%$, and $Se = 99.74\%$ and $P+ = 99.89\%$ for the detection of P-, QRS-, and T-waves, respectively, and the obtained results, which were based on the MIT-BIH arrhythmia database, showed $Se = 99.81\%$ and $P+ = 99.70\%$ for the detection of the QRS complex. Moreover, it will be shown that the results of the proposed method are reliable for a minimum signal quality value of 70%. According to numerical assessments, 8-ms after the occurrence of R-wave, its location will be identified by the computer code of the proposed algorithm. This parameter is 198-ms and 177-ms for P- and T-waves, respectively.

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1. Introduction

The electrocardiogram (ECG) is a simple and non-invasive test for recording the electrical activity of the heart. This signal has three main waves called P, QRS and T, which represent atrial depolarization, ventricular depolarization, and ventricular repolarization, respectively. These waves have significant characteristic

features such as occurrence time, amplitude, intervals, and waveform, and several arrhythmias can be diagnosed on the basis of these features. In any application, in which the instantaneous status of the cardiac system must be monitored, one of the important approaches is to record and analyze the ECG signal in real-time mode. For instance, in the ICU, patients with the risk of cardiogenic shock, hypotensive episodes, sudden cardiac arrest,

Abbreviations: DWT, discrete wavelet transform; db, Daubechies wavelet family; ALC, arc length curve; ASC, area under the slope curve; ASC_f, ASC filtered curve; MSC, mean slope curve; ALC(i^*), amplitude of ALC at the location of the local extremum; $d[n]$, ECG first-order derivative; ASC(i^*), amplitude of ASC at the location of the local extremum; MSC(i^*), amplitude of MSC at the location of the local extremum; A_{QRS} , ASC amplitude at the location of the QRS complex estimated on the basis of beginning of the signal; ALC_m, ALC amplitude average value at the location of R-waves; ASC_m, ASC amplitude average value at the location of R-waves; MSC_m, MSC amplitude average value at the location of R-waves; ALC_R, ALC amplitude at the location of a detected R-wave; ASC_R, ASC amplitude at the location of a detected R-wave; MSC_R, MSC amplitude at the location of a detected R-wave; τ_A , threshold of ASC; C_T , updating waveform template; C_L , waveform template; γ , cross-correlation; C , correlation coefficient; τ_{min}^{AL} and τ_{max}^{AL} , thresholds of acceptable arc length; τ_{min}^N and τ_{max}^N , thresholds of acceptable number of local extremums; AL_L , arc length of a window of length L ; N_{mL} , number of local extremums in a window of length L ; T_T , T-wave template; T_{UT} , updated T-wave template; Se , sensitivity; $P+$, positive predictivity; Sp , specificity; MRT , mean response time; $RMSE$, root mean square error; QTDB, QT database; MITDB, MIT-BIH arrhythmia database; EECF, enhanced external counter pulsation

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and deep unconsciousness during anesthesia must be continuously monitored by evaluating RR-interval variations, T-wave alternans, P-wave variability, QT interval, and ST-segment elevation level. As another example, in heart rehabilitation systems such as enhanced external counter pulsation, the pneumatic control system receives direct feedback from instantaneous locations of R- and P-waves of the patient, which should be provided by a precise real-time ECG analysis algorithm. On the other hand, in implantable devices such as pacemakers and defibrillators, intercardiac ECG signals should be analyzed in real-time with an algorithm of high accuracy with detection of events conducted with low delay. Therefore, providing a fast and accurate algorithm is helpful for extracting ECG signal features and characteristic events. Until now, various methods have been introduced for the real-time detection of the QRS complex, such as methods based on non-linear transform [1], pattern recognition [2], digital filters and Hilbert transform [3–11], wavelet transform [12–14], QRS complex slope estimation [15], adaptive filters [16], and curve length concept [17]. Furthermore, several proposed methods for the real-time detection of P- and T-waves are established on the basis of P- and T-wave slopes [18], Bayesian inference [19], wavelet transform [20], template matching [21], and threshold-based methods [22].

In several methods of the aforementioned researches, two important parameters of real-time analysis algorithms, i.e., “accuracy” and “algorithm response time”, are not considered simultaneously during the design procedure. In other words, reporting only the accuracy of a real-time ECG analysis algorithm is not sufficient and its response time should be considered during design of the method.

In addition, operating characteristics of an algorithm are strongly dependent on the quality of the processed signal. This means that the quality level of the analyzed signal should be reported for which the accuracy of the method is evaluated. So far, this problem is considered in a few previous studies [23,24].

The main goal of this study is to present a new method to improve the response time without loss of accuracy during noisy conditions. In the proposed method, real-time detection of the QRS complex is conducted by applying signal first-order derivative and adaptive threshold adjustment. According to information theories, decision-making based on multiple criteria significantly reduces the false detection of the QRS complex. Moreover, algorithm response time is enhanced desirably with the replacement of high-delay digital filters by efficient low-latency preprocessing algorithms. Correlation analysis conducted between characteristic templates (P- and T-wave templates) and the ECG signal is applied for the online detection of ECG non-impulsive waves. Besides, the algorithm tests signal quality online, which in low-quality cases can estimate the locations of P- and T-waves on the basis of the previously detected locations of characteristic events. The performance of the proposed method is evaluated by its application to the MIT-BIH arrhythmia database (MITDB) [25], QT database (QTDB) [26,27], and Physionet/CinC challenge 2011 competition database [28]. For evaluation of the performance of the proposed method against reduction of signal quality, the algorithm is tested on the open-source artificially generated ECG signal [29,30], whose quality is adjustable according to the random-walk noise theory.

2. Materials and methods

2.1. Datasets

In this study, for performance evaluation of the proposed algorithm, four ECG databases are used. These databases are (1)

the MIT-BIH Arrhythmia Database [25] (48 30-min and two-channel recordings with sampling frequency of 360 Hz), (2) QT Database [26,27] (105 15-min two-channel recordings with sampling frequency of 250 Hz), (3) Physionet/CinC Challenge 2011 [28] (which includes 1500 10-s annotated standard 12-lead ECG recordings with sampling frequency of 500 Hz) and (4) the artificially generated ECG signal [23,29,30] (which includes arbitrary number of quality-variable beats with a tunable sampling frequency). In the third database, the training set is called set “a” (1000 12-lead signals) and the test set is called set “b” (500 12-lead signals). All $1000 \times 12 = 12,000$ signals of the training set (set “a”) are put in a new set called “c”).

2.2. General structure

The general block diagram of the proposed method is presented in Fig. 1.

The proposed method has four major steps:

2.2.1. Real-time preprocessing

For every new sample, the last signal window of a specific length is preprocessed by discrete wavelet transform (DWT). In this process, the powerline interference, high frequency noise, and baseline wander are filtered.

2.2.2. Real-time QRS complex detection

With every new sample, three features are extracted on the basis of the signal and its first-order derivative. Next, QRS complexes are detected by devising appropriate decision rules in which required thresholds are tuned adaptively.

2.2.3. Training phase

After gathering data for 60 s, the training phase begins. This phase includes offline analysis procedures. First, all P- and T-waves are detected. Then, all waveforms are clustered and the characteristic templates are generated by implementing ensemble averaging of members in each cluster [31–33].

2.2.4. Test phase

This phase includes online analysis procedures. With every new sample, first, the quality of the last signal window of a specific length is tested. If the quality level is acceptable, then P- and T-waves are delineated by correlation analysis conducted between the signal and their templates. On the other hand, if the quality of the analysis is window is unacceptable, the locations of the P- and T-waves are estimated on the basis of the locations of previously detected P- and T- waves, respectively. It must be noted that the training phase is continued parallel to the test phase so as to generate the ability of self-tuning when new waveforms appear in the signal. In other words, the templates are updated by detecting all new P- and T-waves and if the correlation between the signal and the templates is lower than a specific threshold, a new cluster is generated.

2.3. Real-time preprocessing

DWT is applied for real-time preprocessing of the signal. This transformation has many advantages such as applicability for analysis of non-linear and non-stationary signals such as ECG, acceptable performance even in short-time frames, and low information destruction of the original signal. Daubechies 4 (db 4) is chosen as the type of the mother wavelet. The original signal is decomposed into 8 levels. First- and second-level detail coefficients and last level approximate coefficients are set to zero to remove high frequency noise and baseline wander effects,

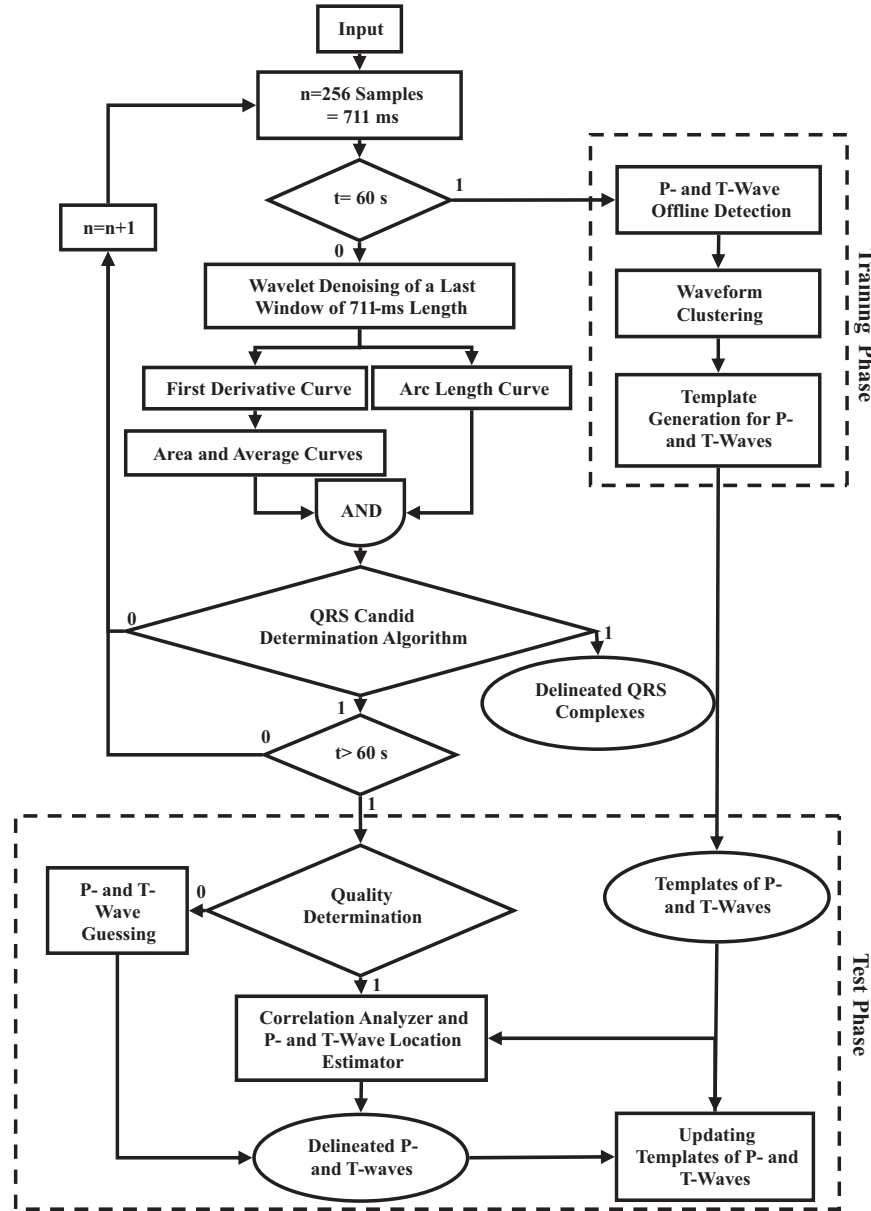


Fig. 1. The general block diagram of the proposed real-time P-QRS-T detection-delineation algorithm.

respectively. The remaining detail coefficients are filtered on the basis of the threshold value introduced by Donoho and Johnstone [34] and soft thresholding function. The denoising procedure is used in an analysis window of approximately 700-ms length. For implementation of the aforementioned denoising, by registering every new sample, a window of approximately 700-ms length, whose end sample is the latest output of the acquisition system, is filtered and reconstructed. The last sample of the reconstruction process is saved as the equivalent filtered value of the last acquisition sample. It should be noticed that this procedure is repeated for every new sample acquisition. However, the samples of the moving window are chosen from the originally acquired samples. In other words, each sample is filtered only once and its filtered value is pushed back to the filtered array.

2.4. Real-time QRS complex detection

The proposed QRS detection algorithm has two main stages including preprocessing and decision rules discussed in the following. All window lengths are determined experimentally so

as to achieve the best possible accuracy and lowest latency for the designed structure.

2.4.1. Preprocessing stage

In the preprocessing stage, three features are extracted from the signal and its first-order derivative. Different methods exist for the calculation of the first-order derivative. The simplest one is the conventional first-order differentiation operator, which causes extreme noise amplification. To prevent noise amplification, the method presented in [31] has been implemented for calculating the first-order derivative of the signal. Because of the averaging process proposed in [31], the noise effect is sufficiently reduced. The first-order derivative is calculated according to Eq. (1):

$$d(i) = \frac{1}{L} \sum_{k=1}^L \tan(\theta_{ik}) \quad (1)$$

where θ_{ik} represents the angle between the line, which connects the i and k samples, and the horizontal line, and L is the window length. The window length is considered to be approximately 27-ms.

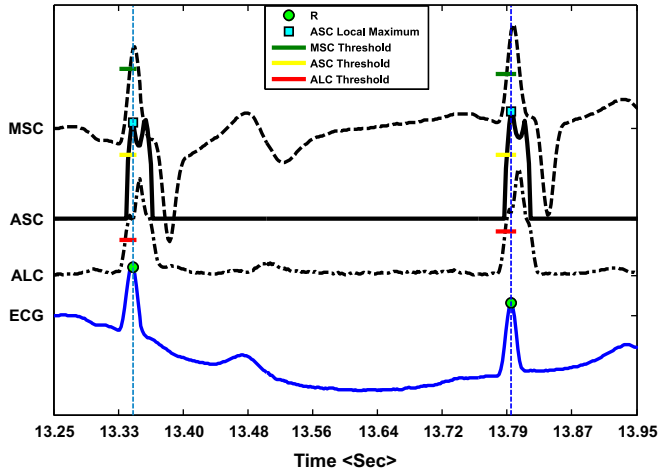


Fig. 2. The three features and their thresholds levels at the location of two QRS complexes.

The features are calculated as follows:

2.4.1.1. Arc length curve (ALC). The first feature is the signal arc length. With every new sample, in the last window of 27-ms length, the signal arc length is determined according to the method presented in [35]. This length is considered as approximately half of the average duration of a QRS complex (normal duration of a QRS complex is 0.06–0.1 s [36]). As a consequence, a local peak value is generated in the ALC curve when every R-wave occurs (Fig. 2).

2.4.1.2. Area under the slope curve (ASC). The second feature is area under the first-derivative curve. With every new sample, according to the method proposed in [35], the area under the first-derivative curve is computed in the last window of 13-ms length. The length of this window is approximately a quarter of the duration of a QRS complex for preventing large delays. Up to this stage, the total delay is composed of the delay caused by the determination of first-order derivative and the delay of the area under curve calculation.

Then, this value is compared with a specific threshold level according to Eq. (2).

$$ASC_{\tau}(i) = \begin{cases} ASC(i) & ASC(i) \geq \tau_A \times A_{QRS} \\ 0 & ASC(i) < \tau_A \times A_{QRS} \end{cases} \quad (2)$$

where $ASC(i)$ is the ASC amplitude at the last registered sample and $ASC_{\tau}(i)$ is its filtered value. Furthermore, A_{QRS} is the ASC amplitude at the location of the QRS complex estimated according to the first 5 s of the signal (in this duration, at least one QRS complex will definitely occur).

τ_A is the ASC threshold that is equal to 0.7, and the selection procedure of this threshold will be discussed in Section 3.1. This feature provides information about the energy variations of the QRS complexes (Fig. 2).

2.4.1.3. Mean slope curve (MSC). The third feature is the mean of the signal first-order derivative in a window of 55-ms length. For MSC calculation, the behavior of the obtained curve will be smoother for a longer window length. Because the locations of Q- and S-waves are detected on the basis of MSC, it should be relatively smoother than other feature signals. Thus, a longer window length is preferred for the extraction of the MSC feature.

This quantity provides information about each QRS complex width.

In Fig. 2, the three feature signals at the location of two QRS complexes are shown.

Then, these features are used in the structure of decision rules in order to detect QRS complexes.

2.4.2. Decision rules stage

2.4.2.1. R-wave detection. To detect R-waves, two conditions are assessed. If these two conditions are met, a new R-wave is detected. The required conditions are as follows:

- A local maximum occurs in ASC (Fig. 2).
- The three feature values calculated on the basis of ALC, ASC, and MSC exceed the following adaptive thresholds at the location of the ASC local maximum (Fig. 2).

$$\begin{cases} ALC(i^*) \geq \alpha_1 \times ALC_m \\ ASC(i^*) \geq \alpha_2 \times ASC_m \\ MSC(i^*) \geq \alpha_3 \times MSC_m \end{cases} \quad (3)$$

where $ALC(i^*)$, $ASC(i^*)$, and $MSC(i^*)$ are ALC, ASC, and MSC amplitudes at the location of the occurred ASC local maximum, and ALC_m , ASC_m , and MSC_m are the averages of ALC, ASC, and MSC quantities at the location of detected R-waves, respectively. (These values are updated with every new R-wave detection.) Moreover, α_1 , α_2 , and α_3 are regulating factors updated by every new QRS complex detection.

The regulating factor for each feature is updated on the basis of the ratio between the detected R-wave feature value to the average of the previous R-wave feature value. Namely, first, the following ratios are computed for features ALC, ASC, and MSC:

$$\begin{cases} \beta_1 = \frac{\alpha_1 \times ALC_m}{ALC_R} \\ \beta_2 = \frac{\alpha_2 \times ASC_m}{ASC_R} \\ \beta_3 = \frac{\alpha_3 \times MSC_m}{MSC_R} \end{cases} \quad (4)$$

where ALC_R , ASC_R , and MSC_R are amplitudes of the three feature signals at the location of the new detected R-wave.

Second, by substituting the value of β_i into the next experimentally obtained function, the regulating factors (α_i s) are determined as follows:

$$\alpha_i = -0.5\beta_i^3 + 0.15\beta_i^2 - 0.015\beta_i + 0.7005, \quad i = 1, 2, 3 \quad (5)$$

The curve of regulating factors α_i versus the β_i value is depicted in Fig. 3.

As represented in Fig. 3, the curve has three main portions:

Part I. In this part, β_i , for $i=1,2,3$, has a small value. It indicates that the threshold level is considerably lower than the feature signal amplitude. Consequently, high frequency noise and P- and T-waves might be falsely detected as R-wave. Therefore, the regulating factor α_i , for $i=1,2,3$, causes increment of the associated threshold level (Eq. (3)) relative to its previous value (Eq. (3)).

Part II. In this part, the thresholds are optimum. In other words, they are large enough to prevent the false detection of noise as R-wave. Meanwhile, they are small enough to detect low-energy R-waves. Therefore, the threshold level will not change significantly in this region.

Part III. In this part, β_i , for $i=1,2,3$, has a large value. It means that the threshold level is around the amplitudes of the feature signal because of which low-energy QRS complexes might not be recognized. Therefore, the regulating factor (α_i , for $i=1,2,3$) causes decrement of the associated threshold level (Eq. (3)) relative to its previous level.

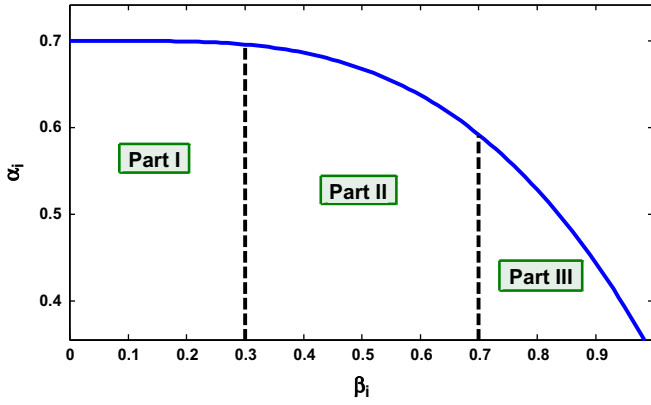


Fig. 3. Determination of R-wave-regulating factors.

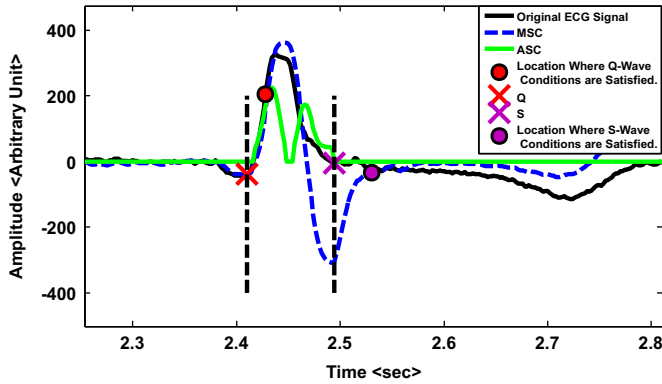


Fig. 4. Q- and S-wave detection criteria. The first location that satisfies Q-wave conditions is the red circle [at this location, Q-wave conditions are satisfied for the first time (Eq. (6))]. The time interval between circles and cross signs shows Q-and S-wave detection latency.

2.4.2.2. Q-wave detection. Q-wave detection is independent of R detection. To detect this wave, the following conditions are checked for every new sample:

$$\begin{cases} \text{ALC}(i) \geq \alpha_1 \times \text{ALC}_m \\ \text{ASC}(i) \geq \alpha_2 \times \text{ASC}_m \\ \text{MSC}(i) \geq \alpha_3 \times \text{MSC}_m \end{cases} \quad (6)$$

where $\text{ALC}(i)$, $\text{ASC}(i)$, and $\text{MSC}(i)$ are the ALC, ASC, and MSC amplitudes of the last registered sample. If these three conditions are satisfied simultaneously, the last MSC local extremum before the i -th sample is marked as Q-wave (Fig. 4).

2.4.2.3. S-wave detection. After the detection of a new R-peak, the first local extremum of MSC is S-wave with the condition that the detected local extremum must have a maximum (/minimum) amplitude in a 40-ms interval (Fig. 4).

2.5. Training phase

For the generation of P- and T-wave templates, a training phase begins after 60 s of data gathering. In this phase, P- and T-waves are delineated in offline mode, and then their waveforms are clustered. Finally, a template is extracted for each cluster by the application of the ensemble averaging technique to its members.

2.5.1. P- and T-wave offline detection in the training phase

Offline detection of P- and T-waves is accomplished by the method presented in [24]. In the presented method, a pattern

recognition algorithm is designed for distinguishing various ECG clusters, which occur between consecutive R- waves. Then, the templates of dominant events of each cluster waveform are extracted. Finally, by the application of correlation analysis, P-and T-waves are delineated (more details are presented in [24]).

2.5.2. P- and T-wave clustering

The criterion for clustering P- and T-waveforms is based on correlation analysis between the obtained waves. Namely, if x is a cluster core sequence, $\tilde{x}$ is a test wave sequence, and $\hat{x}$ is the sequence of difference between them ($\hat{x} = x - \tilde{x}$), then the correlation coefficient between the two waves is computed as follows (proposed in [28]):

$$C = \text{Max} \left(1 - \frac{\int \tilde{x}^2 dt}{\int (x^2 - \text{mean}(x)^2) dt}, 0 \right) \quad (7)$$

The clustering procedure of P- and T-waves is performed as follows. The first detected wave itself is the core of the first cluster. Then, the correlation between the next wave and the first cluster core is determined according to Eq. (7). If the correlation coefficient is greater than a specific threshold, then the test wave is added to this cluster. Otherwise, the test wave itself creates a new cluster. For other waves, this process is repeated.

2.5.3. Extraction of characteristic templates

For the generation of a template for each cluster, the first member is considered as C_T (Fig. 5-a). Then, the following process is repeated separately for all members of the cluster:

- 1- The cross-correlation between the i -th member (W_i) and C_T is determined as follows (Fig. 5-a):

$$\gamma(\tau) = \frac{E[(C_T^t - \mu_{C_T})(W_i^{t+\tau} - \mu_{W_i})]}{\sigma_{C_T} \sigma_{W_i}} \quad (8)$$

where σ_{C_T} and μ_{C_T} , and σ_{W_i} and μ_{W_i} represent the mean value and standard deviation of C_T and W_i , respectively, and τ is the time lag (Fig. 5-b).

- 2- W_i and C_T are aligned at the time lag τ (τ is denoted in Eq. (8)) where they have maximum correlation (Fig. 5-c), and then, their average is calculated as follows (Fig. 5-d):

$$C_L = \frac{C_T \times n + W_i}{n + 1} \quad (9)$$

where n is the number of members incorporated into template extraction until this stage.

- 3- The value of C_T is updated by C_L .

Finally, when this process was repeated for all members, the final C_L is the generated cluster template.

2.6. Test phase

2.6.1. Signal quality analysis

Sometimes, the analysis of the ECG signal is accompanied by significant errors generated either by the chaotic behavior of the heart system (such as arrhythmias, mechanical anomalies, and fibrillations) or by noise potentials, artifacts, and external disturbances. In these situations, chaotic/low-quality segments are generated in the ECG trend; their quality is not sufficient for the application of any analysis algorithm. Hence, even utilization of denoising techniques for the recovery of signal information does not help, and usually, the algorithm recognizes the locations of P- and T-waves falsely. Some examples of different chaotic/low-quality segments that occur in the ECG signals are shown in Fig. 6.

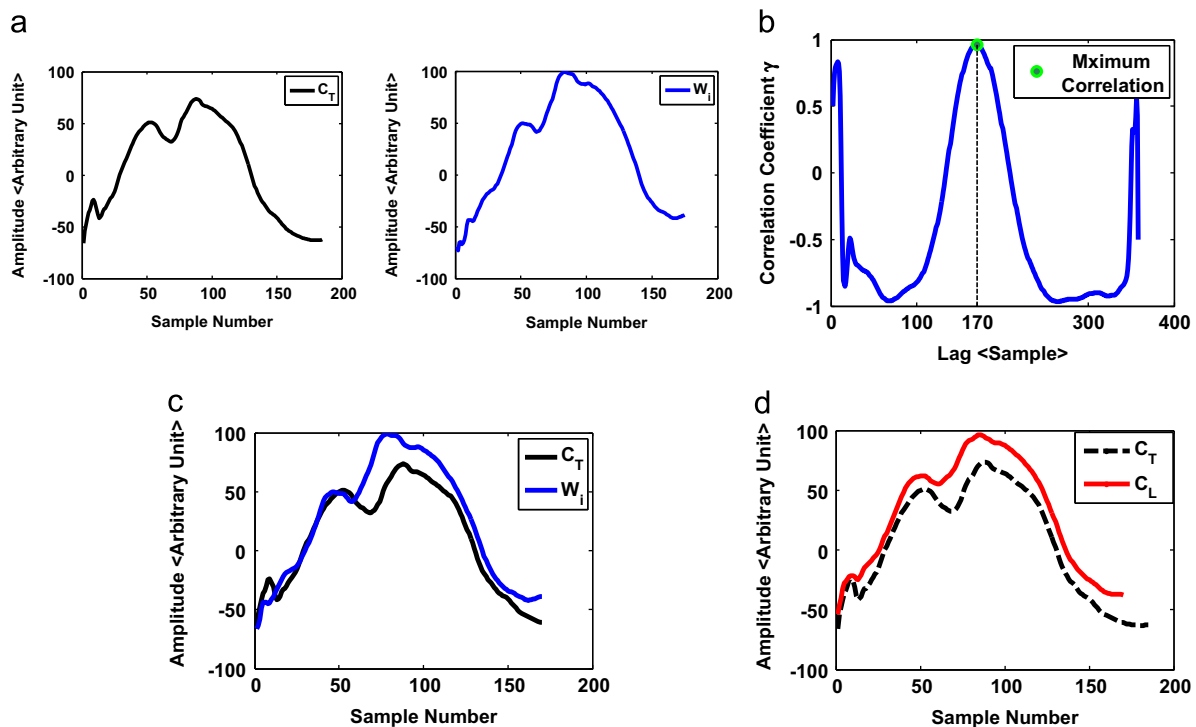


Fig. 5. Procedure of template generation. (a) C_T and W_I , (b) cross-correlation between C_T and W_I and the location where the maximum correlation appears, (c) aligned waves, and (d) the updated template waveform.

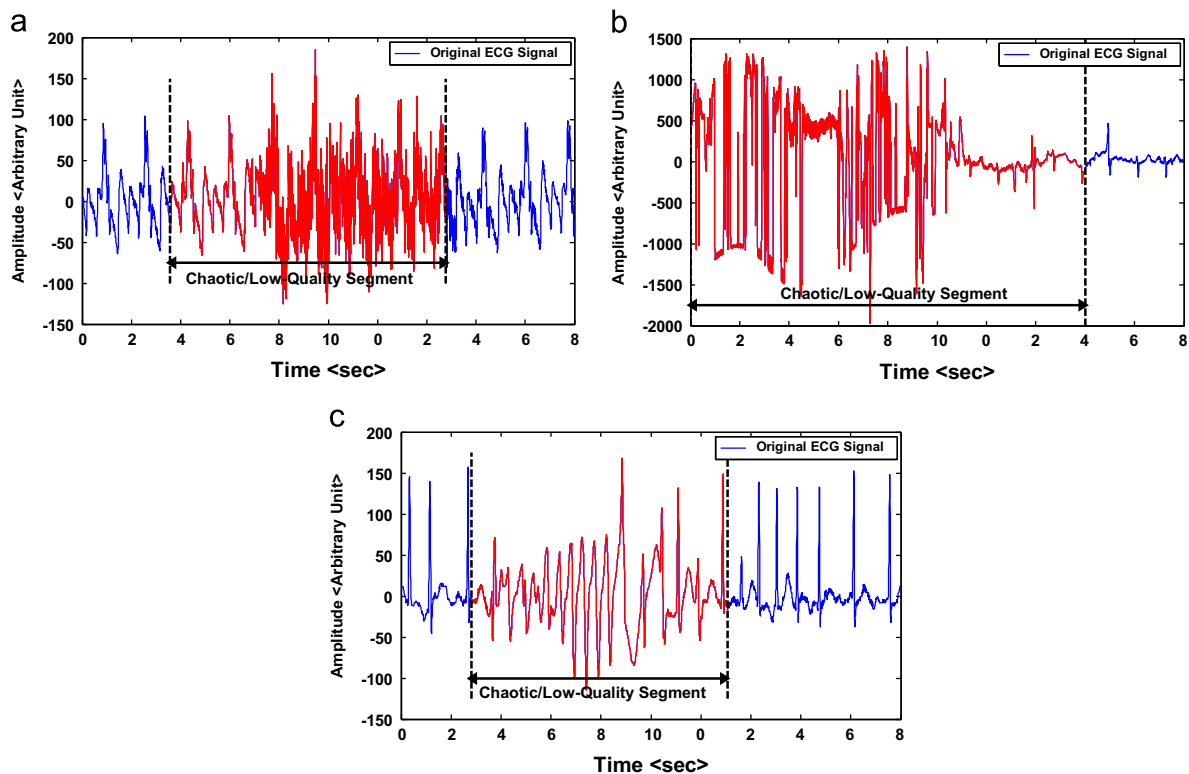


Fig. 6. Examples of different chaotic/low-quality segments in ECG signals.

In such situations, a better option is to guess the locations of P- and T-waves by resorting to previously detected waves. As a result, devising an efficient criterion for the identification of chaotic/low-quality segments is necessary.

Two criteria are employed to distinguish between low- and high-quality parts of the signal. The first one is the number of local extremums, and the second is the arc length in a window of a specific length.

- **Local extremums**

With every new sample, in the last window of 2s length, the number of local extremums (N_{ml}) is computed[31].

- **Arc length**

With every new sample, arc length (AL_L) is calculated according to the method proposed in [35] in the last window of 2s length.

Now, according to Eq. (10), if the number of local extremums and arc length of the analysis window are within the specified range of thresholds, the analysis window has an acceptable quality. On the other hand, if at least one of them is not within the associated admissible range, the quality of the window is unacceptable and the locations of P- and T-waves will be guessed in this window. The mentioned thresholds are determined by histogram analysis, which will be discussed in Section 3.1.

$$\text{Window quality} = \begin{cases} \tau_{\min}^N \leq N_{ml}(i) \leq \tau_{\max}^N \\ \& \\ \tau_{\min}^{AL} \leq AL_L(i) \leq \tau_{\max}^{AL} \end{cases} \quad \text{Acceptable}$$

$$\text{or} \quad \begin{cases} N_{ml}(i) > \tau_{\max}^N \text{ or } N_{ml}(i) < \tau_{\min}^N \\ \text{or} \\ AL_L(i) > \tau_{\max}^{AL} \text{ or } AL_L(i) < \tau_{\min}^{AL} \end{cases} \quad \text{Unacceptable}$$

(10)

where $\tau_{\max}^N$ and $\tau_{\min}^N$ are the upper and lower limits of the acceptable number of local extremums, respectively, and $\tau_{\max}^{AL}$ and $\tau_{\min}^{AL}$ are upper and lower limits of the acceptable arc length, respectively.

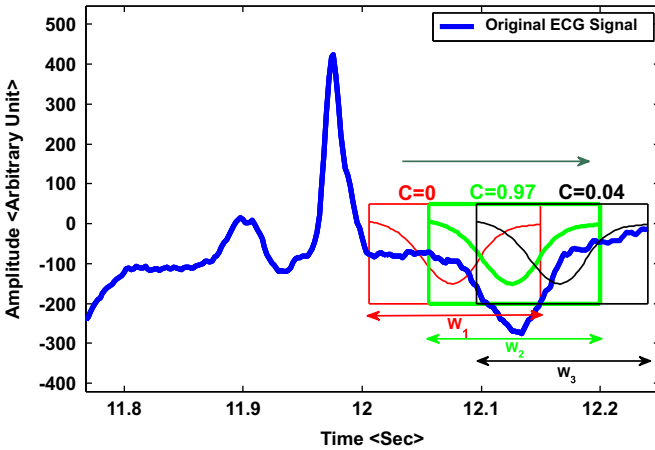


Fig. 7. Forward movement of the sliding window and conduction of the correlation analysis between T-wave templates and each frame.

2.6.2. T-wave real-time detection

After the detection of a new QRS complex, the search for the detection of a new T-wave is started.

With the registration of every new sample, the analysis window is moved one step forward (Fig. 7). Then, the window quality (Eq. (10)) and the correlation coefficient between the analysis window and all of the obtained T-wave templates are computed (Eq. (7)). According to the obtained values, three different cases can occur.

2.6.2.1. Acceptable signal quality and existence of at least one window with a correlation coefficient higher than 0.8. The window that is more correlated with T-wave templates among others is T-wave. The search for this wave is terminated until beginning of the new cycle. Furthermore, the cluster whose template has the highest correlation with the T-wave templates is the winner cluster. A new member is added to the winner cluster, and the template of this cluster is updated as follows:

$$T_{UT} = \frac{T_T \times n + T_N}{(n+1)} \quad (11)$$

where T_N is the new detected T-wave, T_T is the winner cluster template, n is the number of members in the cluster, and T_{UT} is the updated cluster template.

2.6.2.2. Acceptable signal quality and no window with a correlation coefficient higher than 0.8. In the training phase, the interval between each R-wave and the next T-wave offset is estimated. If a window with a correlation coefficient higher than 0.8 is not observed in the interval following the occurrence of the new R-wave, then a new waveform is considered to have occurred. In this situation, the algorithm estimates the location of T-wave by utilizing the information of previously detected T-waves; the new wave itself constitutes a new cluster.

2.6.2.3. Unacceptable quality. In this case, the location of T-wave is estimated by utilizing the information of previously detected T-waves. However, this wave cannot constitute a new cluster.

These three probable cases are shown in Fig. 8.

2.6.3. Real-time P-wave detection

In general, real-time P-wave detection is similar to the T-wave detection process. However, there are two noticeable differences mentioned below:

- 1- Search for P-wave begins after lapse of $0.6 \times RR_{\text{average}}$ [36] (RR_{average} is the average of previous RR intervals, which is updated after the detection of every new R) from the new R-wave.

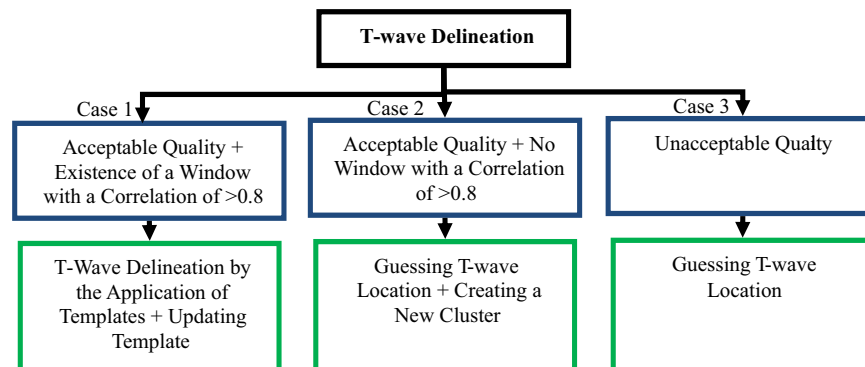


Fig. 8. Three different possibilities in real-time T-wave detection.

- 2- Absence of P-wave (this case could occur in different arrhythmias such as atrial fibrillation, sinoatrial block, and junctional rhythm [36]) is also checked by the algorithm.

2.6.3.1. *Absent P-wave.* If there was no window with a correlation coefficient higher than 0.8, the algorithm does not make the decision for the location of P-wave till the beginning of the next cycle. Because the maximum duration of PR interval is approximately 200-ms [36], if there is no local extremum in this duration, P-wave is assumed to be absent in the search interval. Otherwise, a new waveform can be found and a new cluster will be created. Fig. 9 shows all the probable cases in the real-time detection of P-wave.

3. Results

3.1. Determination of the optimum value for thresholds

All the utilized thresholds and window durations are obtained experimentally so as to achieve the best possible results. In the following section, the procedure of their selection is discussed.

3.1.1. ASC threshold determination (τ_A)

In Section 2.4.1, it is expressed that a threshold is considered for ASC. If the amplitude of this curve is less than $0.7 \times A_{QRS}$ (Eq. (2)), then its value will be set to zero. For the determination of this threshold optimum level, its value is altered from 0.1 to 0.9, and during each increment, the accuracy of the QRS detection algorithm is evaluated. The algorithm is implemented to an artificially generated ECG signal expressed in Section 2.1. The quality of this signal is reduced to 0.7 by the random-walk noise theory.

According to Fig. 10, when the threshold value τ_A is assigned to approximately 0.7, ($\tau_A = 0.7$), the best possible performance can be achieved by the algorithm.

3.1.2. Determining signal quality thresholds ($\tau_{max}^{AL}, \tau_{min}^{AL}, \tau_{max}^N, \tau_{min}^N$)

In Section 2.6.1, four thresholds are expressed for the determination of signal quality. For the determination of these thresholds, a window of 2s length is moved on all the signals of set “c” (Section 2.1). In each frame, the number of local extremums (index1) and the arc length (index2) are calculated. In each signal, index1 of the frame, whose number of local extremums is the greatest among all other frames, is selected as the signal first index. Furthermore, index2 of the frame, whose arc length is the greatest among all other frames, is selected as the signal second index. The distributions of these indices are illustrated in two histograms, (Figs. 11 and 12). As it can be seen in Figs. 11 and 12, the number of extremums and arc length of the most signals lie between specific ranges. Thus, if at least one of the number of extremums or arc length of a signal is located out of the admissible ranges, the signal quality is low and should be categorized as unacceptable. In Section 2.6.1, the associated equations are expressed.

3.2. Performance evaluation of real-time QRS detection

To evaluate the performance of the proposed real-time QRS complex detection method, the algorithm is applied to lead I of all 48 records of MITDB [25] (Table 1). The results of the Pan and Tompkins method are also presented in this table. The two methods are compared according to their three performance characteristics: sensitivity (Se), positive predictivity (P+), and mean response time (MRT).

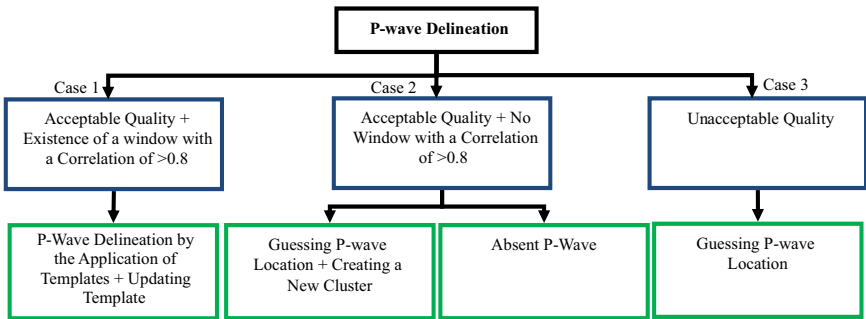


Fig. 9. Different cases in real-time P-wave detection.

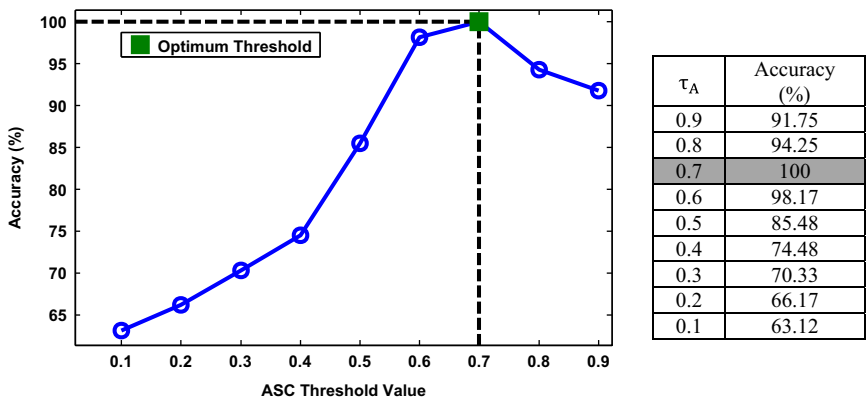


Fig. 10. QRS detection accuracy against different ASC thresholds (Eq. (2)).

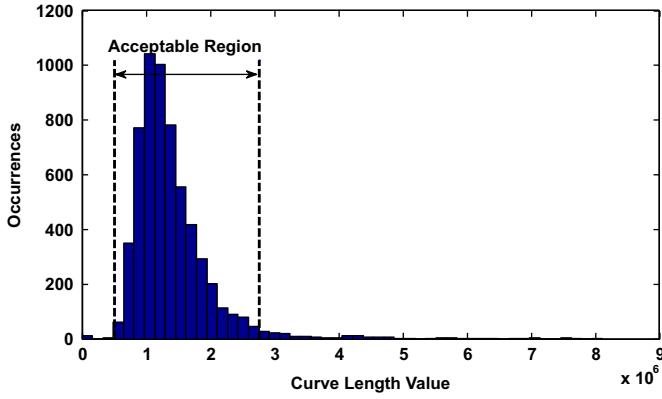


Fig. 11. Distribution of the curve length of the set “c” signals in a window of 2-s length.

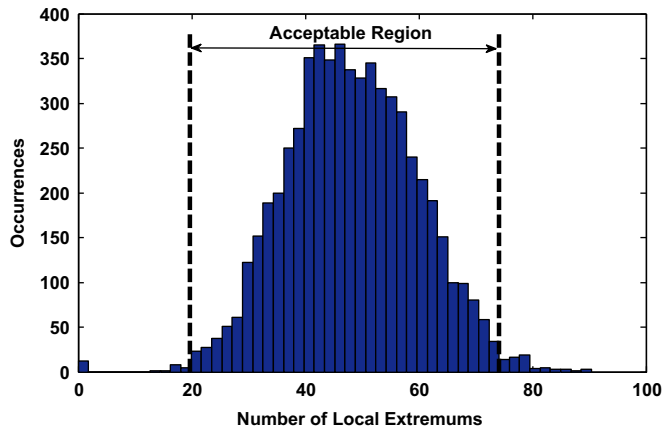


Fig. 12. Distribution of the number of local extremums of the set “c” signals in a window of 2-s length.

Se, P+, and MRT are defined as follows:

$$\begin{aligned} \text{Se} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \% \\ \text{P+} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \% \end{aligned} \quad (12)$$

where TP is the number of true-positive detections, FN is the number of false-negative detections, and FP is the number of false-positives detections.

Moreover, MRT can be defined as follows:

$$\text{MRT} = \frac{1}{n} \sum_{i=1}^n |\text{Detection time}_i - \text{Actual time}_i| \times 1000 \text{ ms} \quad (13)$$

where i represents the i -th detected event and n is the total number of events.

The proposed method for the QRS complex detection are $\text{Se}=99.81\%$ and $\text{P+}=99.70\%$. These operating characteristics are $\text{Se}=99.75\%$ and $\text{P+}=99.53\%$ for the Pan and Tompkins algorithm [10]. Besides, the algorithm MRT for R-wave detection is 8-ms in this study and is 124-ms in the Pan and Tompkins algorithm. The comparison conducted between the proposed method and the methods proposed in some previous studies are presented in Table 2.

In Section 2.4.2, it is expressed that all R-wave-related thresholds are adaptive. The alternation of the ASC threshold (Eq. (3)) in an arbitrary ECG signal is shown in Fig. 13. According to this figure, if a constant threshold is chosen, many of the low-energy QRS complexes are missed falsely.

3.3. Performance evaluation of the signal quality analysis algorithm

To assess the proposed signal quality evaluation method, the method is tested on set “b” (test set) of Physionet/CinC Challenge 2011 competition database [28]. The achieved results are presented in Table 3 in terms of accuracy and specificity (Sp) defined as follows:

$$\text{Sp} = \frac{\text{TN}}{\text{TN} + \text{FP}} \% \quad (14)$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \% \quad (15)$$

In addition, the results of this method are compared with those of some of the other quality evaluation methods in Table 4.

3.4. Performance evaluation of the real-time P-QRS-T detection algorithm

For the evaluation of the real-time P-QRS-T detection method, the algorithm is applied to 40 records of QTDB[26,27] (Table 5). Overall performance of the algorithm in terms of four evaluation parameters is shown in Table 6. The parameter RMSE in this table is calculated according to Eq. (16). This parameter shows the RMS error between the real and predicted locations of the events:

$$\begin{aligned} \text{RMSE} &= \sqrt{\frac{\sum_{i=1}^n (\text{Predicted Event Time}_i - \text{Actual Event Time}_i)^2}{n}} \\ &\times 1000 \text{ ms} \end{aligned} \quad (16)$$

where i is the i -th detected event and n is the total number of events.

Because of the low amplitude and low energy of P-waves and the similarity of some noisy components to this wave, the accuracy of P-wave detection is lower than that of T-wave detection. Moreover, RMSE is calculated to show the average error between the real and predicted locations of the events. This parameter is 15.55-ms, 6.43-ms, and 10.63-ms for P, R, and T peaks, respectively, indicating more complexity of the detection of P-wave relative to that of other waves.

The algorithm MRT for the detection of P- and T-waves is 198-ms and 177-ms, respectively. Opposite to the detection of QRS complexes, the detection of P- and T-waves can be completed only after the complete occurrence of these waves. Thus, this process causes a considerably more delay in detecting P- and T-waves than in detecting R-waves. However, a compromise exists between algorithm accuracy and MRT in the detection of P- and T-waves. It means that as far as response time increases, the algorithm detects the locations of P- and T-waves with greater reliability. On the other hand, reducing response time causes an increase in the probable false-positive/false-negative errors of the algorithm.

Table 7 shows a comparison between the performance of the presented method with the methods presented in some previous studies.

3.5. Performance evaluation of the proposed method against different signal qualities

In this study, the robustness of the proposed method in maintaining the required precision under reduction of signal quality is evaluated. To this end, the algorithm is tested on the artificially generated ECG signal mentioned in Section 2.1. The quality of this signal is altered from the best case (100%) to the worst case ($\leq 40\%$) according to the random-walk theory (Fig. 14). For example, a 40% quality means that the approximate correlation

Table 1

Results obtained by the application of the proposed QRS detection method to MITDB along with comparison with the Pan and Tompkins algorithm [10].

Tape (No.)	Total (No. of beats)	FP (No. of beats)	FN (No. of beats)	Failed detection this study	Failed detection Pan and Tompkins	MRT (ms) this study	MRT (ms) Pan and Tompkins	Se (%) this study	P+ (%) this study	Se (%) Pan and Tompkins	P+ (%) Pan and Tompkins
100	2273	0	0	0	0	8	128	100	100	100	100
101	1865	1	3	0.21	0.43	8	133	99.84	99.73	99.84	99.95
102	2187	0	0	0	0	8	121	100	100	100	100
103	2084	0	0	0	0	7	110	100	100	100	100
104	2230	2	5	0.31	0.04	8	124	100	99.96	99.78	99.91
105	2572	52	10	2.41	3.46	8	123	99.14	97.44	99.61	98.01
106	2027	0	9	0.44	0.34	8	106	99.90	99.75	99.56	100
107	2137	0	0	0	0.09	8	133	99.91	100	100	100
108	1763	145	73	12.40	12.54	8	124	98.75	89.74	95.86	92.10
109	2532	0	2	0.08	0.04	8	111	99.96	100	99.92	100
111	2124	1	0	0.05	0.05	8	111	100	99.95	100	99.95
112	2539	0	0	0	0.04	8	101	99.96	100	100	100
113	1795	0	0	0	0	8	120	100	100	100	100
114	1879	3	0	0.16	1.06	8	138	99.10	99.84	100	99.84
115	1953	0	0	0	0	8	127	100	100	100	100
116	2412	2	7	0.37	1.04	8	107	99.09	99.87	99.71	99.92
117	1535	1	0	0.07	0.13	8	109	99.93	99.93	100	99.93
118	2275	1	0	0.04	0.04	8	139	100	99.96	100	99.96
119	1987	0	1	0.05	0.05	8	136	100	99.95	99.95	100
121	1863	1	5	0.32	0.59	8	119	99.62	99.78	99.73	99.95
122	2476	0	0	0	0.08	8	133	99.96	99.96	100	100
123	1518	0	0	0	0	8	116	100	100	100	100
124	1619	0	0	0	0	8	139	100	100	100	100
200	2601	3	2	0.19	0.35	8	124	99.88	99.77	99.92	99.88
201	1963	0	7	0.36	0.51	8	127	99.49	100	99.64	100
202	2136	0	0	0	0.19	8	136	99.81	100	100	100
203	2982	71	36	3.59	2.87	8	128	98.99	98.24	98.79	97.65
205	2656	0	0	0	0.08	8	127	99.92	100	100	100
207	1862	5	2	0.38	0.43	8	126	99.79	99.79	99.89	99.73
208	2956	0	7	0.24	0.60	8	131	99.53	99.86	99.76	100
209	3004	0	0	0	0.10	8	130	100	99.90	100	100
210	2647	1	5	0.23	0.38	8	122	99.70	99.92	99.81	99.96
212	2748	0	0	0	0	8	128	100	100	100	100
213	3251	0	1	0.03	0.09	8	129	99.94	99.97	99.97	100
214	2262	2	2	0.18	0.26	8	127	99.82	99.91	99.91	99.91
215	3363	0	0	0	0.03	8	110	99.97	100	100	100
217	2208	0	6	0.27	0.45	8	126	99.73	99.82	99.73	100
219	2154	0	0	0	0	8	123	100	100	100	100
220	2048	0	0	0	0	8	124	100	100	100	100
221	2427	0	1	0.04	0.08	5	130	100	99.92	99.96	100
222	2484	0	0	0	7.33	7	129	96.74	95.97	100	100
223	2605	0	0	0	0.04	5	130	100	99.96	100	100
228	2053	15	8	1.12	1.46	8	116	99.76	98.79	99.61	99.27
230	2256	0	0	0	0.04	6	109	100	99.96	100	100
231	1886	0	0	0	0	8	131	100	100	100	100
232	1780	2	0	0.11	0.39	6	128	99.94	99.66	100	99.89
233	3079	0	0	0	0.03	7	144	99.97	100	100	100
234	2753	0	0	0	0	5	130	100	100	100	100
48	116,137	308	192	0.49	0.675	8	124	Se=99.81	P+=99.70	Se=99.75	P+=99.53

Table 2

Comparison of the proposed real-time QRS detection method and the methods proposed in some previous studies. Application to MITDB.

Method	Se (%)	P+ (%)
This study	99.81	99.70
Hamilton–Tompkins [9]	99.68	99.63
Pan–Tompkins [10]	99.75	99.56
Chen et al. [8]	99.59	99.49
Arzeno et al. [3]	98.07	99.18

between the original clean ECG and its noisy version is approximately 0.4 (more details are presented in [23]). In each decrement, the algorithm accuracy is evaluated (Table 8). Low signal quality leads to a lower correlation between the signal and the obtained templates. Therefore, several new clusters will be created and consequently, guessing the locations of P- and T-waves will

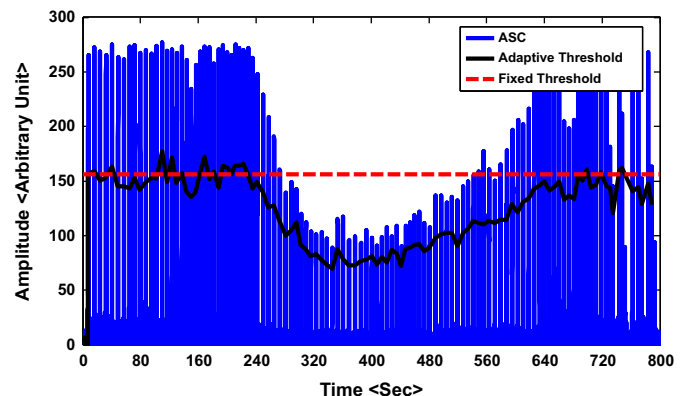


Fig. 13. Alternation of the ASC threshold (Eq. (3)) in an arbitrary ECG signal. The fixed threshold is set by the algorithm initially based on the first 5 s of the signal. Record 106 of MITDB.

increase. Hence, the accuracy and reliability of the algorithm will decrease. According to Table 8, when the signal quality is lower than 60%, at least 10% of P- and T-waves detections take place by

the guessing process. In this manner, detection accuracy will decline to lower than 70%. The behavior of algorithm accuracy against the quality value of the signal is shown in Fig. 15.

Table 3

Quality evaluation of 500 test signals of the Physionet/CinC challenge 2011 competition database (set “b”) [28].

Algorithm	TP (No. of beats)	FP (No. of beats)	TN (No. of beats)	FN (No. of beats)	Sensitivity (%)	Specificity (%)	Accuracy (%)
Quality evaluation	2540	291	3225	85	98.51	92.27	93.88

Table 4

Comparison of the proposed signal quality evaluation method with the methods proposed in some previous studies. Application to Physionet/CinC challenge 2011 [28].

Method	Accuracy (%)
This study	93.87
Xia et al. [37]	93.20
Clifford et al. [38]	92.60
Tat et al. [4]	92.00
Hayn et al. [39]	91.60
Di Marco et al. [40]	91.40

Table 5

Implementation of the real-time P-QRS-T detection algorithm to 40 records of QTDB.

Tape	Beats (No.)	FN _P (No.)	FP _P (No.)	RMSE _P (ms)	Se _P (%)	P+P (%)	FN _R (No.)	FP _R (No.)	RMSE _R (ms)	Se _R (%)	P+R (%)	FN _T (No.)	FP _T (No.)	RMSE _T (ms)	Se _T (%)	P+T (%)
30	1051	1	1	14	99.90	99.90	0	0	13	100	100	1	7	0	99.90	99.34
31	1071	7	7	26	99.35	99.35	0	0	0	100	100	7	0	21	99.35	100
32	1250	17	1	13	98.64	99.92	17	0	13	98.64	100	17	2	0	98.64	99.84
33	524	3	3	0	99.43	99.43	0	0	0	100	100	3	3	1	99.43	99.43
34	957	3	3	9	99.69	99.69	0	0	12	100	100	3	1	6.5	99.69	99.9
35	813	5	1	12	99.38	99.88	8	0	11	99.02	100	5	2	12	99.38	99.75
37	626	5	2	21	99.20	99.68	5	0	21	99.20	100	5	0	0	99.20	100
38	1808	0	0	0	100	100	0	0	10	100	100	0	0	1	100	100
39	1356	1	2	12	99.93	99.85	1	0	0	99.93	100	1	2	8	99.93	99.85
40	1017	0	0	21	100	100	0	0	8	100	100	0	0	26	100	100
41	1356	1	2	17	99.93	99.85	1	2	10	99.93	99.85	1	1	15	99.93	99.93
42	1252	2	1	10	99.84	99.92	2	1	9	99.84	99.92	2	0	27	99.84	100
43	1252	1	0	21	99.92	100	0	0	26	100	100	1	0	12	99.92	100
44	1162	4	0	21	99.66	100	1	0	0	99.91	100	10	0	9	99.14	100
45	957	1	1	8	99.90	99.9	1	0	8	99.90	100	1	1	29	99.90	99.9
46	775	2	0	12	99.74	100	2	0	0	99.74	100	2	0	10	99.74	100
47	856	0	0	32	100	100	0	0	0	100	100	0	0	6	100	100
48	1479	2	0	6	99.86	100	0	0	5	100	100	0	1	12	100	99.93
49	814	0	1	51	100	99.88	0	0	0	100	100	3	0	14	99.63	100
50	543	3	0	39	99.45	100	3	0	0	99.45	100	3	0	22	99.45	100
51	739	0	3	0	100	99.60	0	0	10	100	100	0	0	0	100	100
52	1479	2	0	26	99.86	100	0	0	0	100	100	0	0	21	100	100
100	1134	0	0	20	100	100	0	0	16	100	100	0	0	0	100	100
102	1088	12	1	10	98.90	99.91	1	0	0	99.91	100	1	0	0	99.91	100
103	1049	10	0	0	99.05	100	10	0	0	99.05	100	10	0	0	99.05	100
104	1085	4	0	29	99.63	100	4	0	19	99.63	100	4	0	28	99.63	100
114	870	2	1	0	99.77	99.88	0	0	0	100	100	0	0	0	100	100
116	1186	0	0	22	100	100	0	0	1	100	100	0	0	27	100	100
117	766	1	0	0	99.87	100	1	0	0	99.87	100	1	5	0	99.87	99.35
233	1582	12	5	33	99.24	99.68	1	0	12	99.94	100	3	2	4	99.81	99.87
306	1040	6	4	12	99.42	99.61	0	0	3	100	100	0	2	10	100	99.81
310	2012	6	1	13	99.70	99.95	2	0	1	99.90	100	5	3	12	99.75	99.85
803	1026	7	2	16	99.32	99.80	0	1	0	100	99.90	3	0	16	99.71	100
872	990	3	7	10	99.70	99.30	0	0	7	100	100	1	4	6	99.90	99.6
883	893	1	6	8	99.89	99.33	0	0	13	100	100	2	0	8	99.78	100
6795	786	2	0	17	99.75	100	1	0	4	99.87	100	3	6	17	99.62	99.24
14,046	1260	10	9	21	99.21	99.29	2	2	17	99.84	99.84	2	2	3	99.84	99.84
14,157	1081	3	3	8	99.72	99.72	0	1	3	100	99.90	0	0	14	100	100
15,814	1036	7	3	7	99.32	99.71	1	0	4	99.90	100	2	0	19	99.81	100
16,273	1112	10	0	25	99.10	100	0	0	1	100	100	0	3	9	100	99.73
40	43,133	156	70	15.55	99.63	99.83	64	7	6.43	99.83	99.98	102	47	10.63	99.74	99.89

Table 6

Results of implementation of the real-time P-QRS-T detection algorithm to 40 records of QTDB [26,27].

	FP (No. of beats)	FN (No. of beats)	Se (%)	P+ (%)	RMSE (ms)	MRT (ms)
P_{peak}	70	156	99.63	99.83	15.55	198
R	7	64	99.83	99.98	6.43	8
T_{peak}	47	102	99.74	99.89	10.63	177

Table 7

Comparison of the proposed real-time P-QRS-T detection method with the methods proposed in some previous studies.

Method	Operation characteristics	P_{peak}	R	T_{peak}
This study	Se (%) P+ (%)	99.63	99.83	99.74
		99.83	99.98	99.89
Chatterjee et al. [18]	Database Mode	QT		
	Se (%) P+ (%)	Online		
		97.58	NR	97.78
		96.84	NR	98.04
Martínez et al. [41]	Database Mode	PTB [40]		
	Se (%) P+ (%)	Online		
		99.20	99.20	99.20
		99.01	99.01	99.01
Homaeinezhad et al. [24]	Database Mode	QT		
	Se (%) P+ (%)	Offline		
		99.97	99.97	99.97
		99.99	99.99	99.99
Madeiro et al. [42]	Database Mode	QT		
	Se (%) P+ (%)	Offline		
		99.32	99.32	99.32
		99.47	99.47	99.47
	Database Mode	QT		
		Offline		

NR: Not Reported NA: Not Applicable.

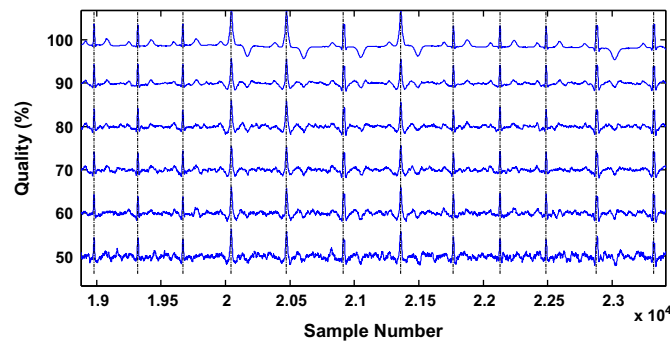


Fig. 14. Altering the artificial ECG signal quality based on the random-walk noise theory.

Table 8

Performance of the proposed method against different signal qualities.

Quality (%)	R			P_{peak}			T_{peak}		
	Accuracy (%)	RMSE (ms)	Guessed ^a (%)	Accuracy (%)	RMSE (ms)	Guessed ^a (%)	Accuracy (%)	RMSE (ms)	Guessed ^a (%)
100	100	6	0	100	17.22	0	100	12.10	0
90	100	6	0	100	21.02	0	100	16.20	0
80	100	6	0	92.06	25.87	4	97.67	17.23	0.34
70	100	8	0	83.95	36.70	7.67	88.68	24.10	4.34
60	100	10	0	62.16	44.10	16.34	70.45	35.80	9.67
50	100	9	0	42.86	52.60	27	60	43.60	14.67
40	99.34	23	0	36.36	68.90	33.67	51.13	51.01	26

^a Guessed: This parameter shows the percentage of events whose locations are estimated based on locations of previously detected events.

4. Discussion

In real applications, because the algorithm might face new situations at any time, real-time ECG signal analysis has great difficulties. The proposed method benefits the main feature that it has ability to adapt itself to any new condition. Clustering P- and T-waveforms and generating new clusters under forthcoming conditions significantly increase the accuracy of the algorithm such that the results of the proposed real-time method are quite comparable with those of other offline methods [24,41,42].

Taking advantages of wavelet denoising causes a strong reduction in the noise impact of the signal, which eventuates notable decrement in FP detection in comparison to FN detection during the identification of P-QRS-T waves (Table 5).

QRS complex detection is the basic step for P- and T-wave delineation. Therefore, higher the accuracy and lower the latency of QRS complex detection, greater the speed and accuracy of P- and T-wave delineation. In the proposed QRS detection algorithm, considering short length windows decreases the algorithm response time significantly. In the previously proposed methods such as [10], utilization of digital filters with a large length results in imposition of high delays in the detection of the QRS complex. The response times of the proposed algorithm and the method presented in [10] are approximately 8-ms and 124-ms, respectively. On the other hand, multi-criteria decision rules based on the amplitude and first-order differentiation of the signal increase the accuracy along with a decrease in the detection of the RMS error.

5. Conclusion

In this study, a method is proposed for the real-time detection of P-QRS-T waves of the ECG signal. Real-time QRS complex

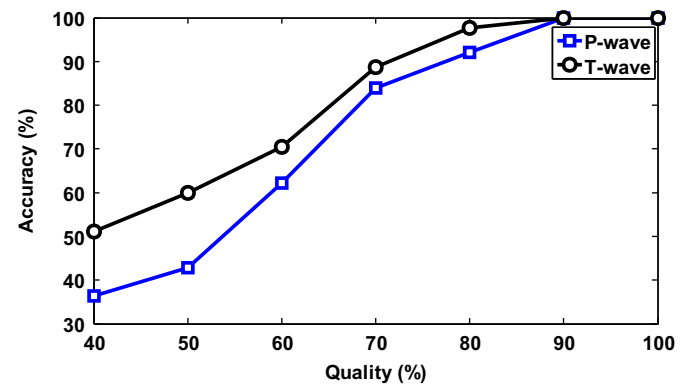


Fig. 15. Variation in P- and T-waves detection accuracy versus the quality value of artificial ECG signal.

detection is performed on the basis of the amplitude and first-order derivative of the signal. The obtained results show high accuracy and short response time of the algorithm. The algorithm detects QRS complexes with $Se=99.83\%$ and $P+=99.98\%$ based on QTDB and $Se=81\%$ and $P+=70\%$ based on MITDB. Moreover, MRT is approximately 8-ms. P- and T-wave real-time detection is based on correlation analysis conducted between their templates and the signal. Based on QTDB, the proposed method has $Se=99.63\%$ and $P+=99.83\%$, and $Se=99.74\%$ and $P+=99.89\%$ for the detection of P- and T- waves, respectively. As an important merit, the algorithm delays for the detection of P- and T-waves are 198-ms and 177-ms, respectively. In addition, it is shown that the algorithm performance is reliable for signal quality of approximately higher than 70%. High accuracy, short response time, and simplicity are the main features of the proposed procedure.

Conflict of interest statement

No conflict of interest.

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